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The role of nitrogen dioxide in the prevalence of adverse cardiovascular and cerebrovascular diseases in China: a national multi-pollutant geospatial analysis

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Abstract

Introduction Cardiovascular and cerebrovascular diseases (CCVDs) pose a severe global health threat, particularly among middle-aged and elderly populations. Ambient air pollution is a well-recognized environmental risk factor for CCVDs, yet existing studies often focus on single pollutants or outcomes, lacking comprehensive evaluations. This study aimed to systematically explore the associations between multiple air pollutants and three core adverse CCVD outcomes (hypertension, heart disease, stroke) in middle-aged and elderly individuals.

Methods and materials Data were integrated from the China Health and Retirement Longitudinal Study (CHARLS) and the China High-resolution Air Pollutants (CHAP) dataset, including seven major air pollutants (PM₁, PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃) and multi-dimensional covariates (demographic characteristics, lifestyle habits, underlying diseases). Two analytical phases were conducted: AQI-related analysis (2015–2020, 20,990 participants) and machine learning analysis (2013–2020, 22,517 participants). Statistical methods included descriptive analysis, univariate linear regression, multivariate logistic regression, restricted cubic spline (RCS) analysis, and six machine learning algorithms (XGBoost, Random Forest, etc.) combined with Shapley Additive Explanations (SHAP) values.

Results City-level univariate linear regression showed significant positive correlations between annual average AQI and the prevalence rates of hypertension ($\beta = 0.0006$, $P = 0.0125$), heart disease ($\beta = 0.0008$, $P = 0.0147$), and stroke ($\beta = 0.0005$, $P = 0.0248$). At the individual level, multivariate logistic regression revealed that each 1-unit increase in AQI was associated with a 3% higher risk of hypertension (OR = 1.003, $P < 0.001$), 6% higher risk of heart disease (OR = 1.006, $P < 0.001$), and 4% higher risk of stroke (OR = 1.004, $P = 0.0433$) after full covariate adjustment. RCS analysis indicated non-linear dose–response relationships, with accelerated risk increases at moderate-to-high AQI levels (≥ 80 for heart disease, ≥ 100 for stroke). Machine learning models exhibited good predictive performance (AUC range: 0.720–0.771),

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and SHAP value analysis identified NO₂ as the most critical pollutant influencing all three outcomes, with significantly higher impact weights than other pollutants.

Discussion This study confirms that long-term air pollution exposure is an independent risk factor for hypertension, heart disease, and stroke in middle-aged and elderly individuals. NO₂ plays a core role through mechanisms such as disrupting vascular endothelial function, inducing oxidative stress, and triggering inflammatory responses. These findings provide a scientific basis for precise environmental health interventions and CCVD prevention, particularly emphasizing NO₂ control.

Conclusion This study confirms that air pollution is an independent risk factor for hypertension, heart disease, and stroke in middle-aged and elderly individuals. Crucially, machine learning analysis identified NO₂ as the dominant pollutant, with a greater impact than particulate matter.

Keywords Air pollution, Nitrogen dioxide (NO₂), Cardiovascular and cerebrovascular diseases (CCVDs), Machine learning

Introduction

Cardiovascular and cerebrovascular diseases (CCVDs), as the primary health threat globally, have emerged as a core component of the health burden among middle-aged and elderly populations [1]. According to statistics, approximately 18 million people die from cardiovascular diseases worldwide each year, among which hypertension affects around 1 billion individuals and stroke patients number about 240 million [2, 3]. With the intensification of population aging, it is projected that by 2030, the proportion of people aged 65 and above will reach 20%, the share of CCVDs as the leading cause of death will rise to 40%, and the associated treatment costs will increase substantially [4].

As an environmental risk factor, the detrimental effects of ambient air pollution on cardiovascular and cerebrovascular health have garnered extensive attention. Nitrogen dioxide (NO₂), a typical gaseous pollutant, can enter the circulatory system through diffusion, directly attacking vascular endothelial cells (VECs), impairing their integrity and functional stability, and inducing cardiovascular and cerebrovascular embolism [5]. NO₂ can also trigger excessive production of reactive oxygen species (ROS) in endothelial cells, promoting the release of inflammatory factors (e.g., TNF- α , IL-6). This further stimulates the proliferation of vascular smooth muscle cells and collagen synthesis, leading to vascular wall thickening, lumen stenosis [6], reduced vascular elasticity, and ultimately the development of hypertension. The harm of carbon monoxide (CO) to the cardiovascular and cerebrovascular systems primarily stems from its interference with oxygen transport mechanisms and subsequent cellular metabolic disorders, resulting in hypoxia in vital organs such as the heart and brain. Hypoxia in cardiomyocytes activates the hypoxia-inducible factor-1 α (HIF-1 α) signaling pathway. Although this can upregulate the expression of vascular endothelial growth factor (VEGF) to promote angiogenesis for oxygen supply compensation, sustained hypoxia leads to excessive activation

of HIF-1 α . This triggers a remodeling of cardiomyocyte energy metabolism from aerobic oxidation to anaerobic glycolysis, causing lactic acid accumulation, decreased pH value, and ultimately inducing cardiomyocyte acidosis and functional damage [7, 8]. For the cerebrovascular system, hypoxia can induce damage to cerebral microvascular endothelial cells, increase blood–brain barrier (BBB) permeability, trigger cerebral edema and inflammatory responses, and simultaneously promote platelet aggregation and thrombosis, significantly elevating the risk of ischemic stroke [9].

Ozone (O₃), a highly oxidizing pollutant, exerts its damaging effects on the cardiovascular and cerebrovascular systems mainly through inducing oxidative stress and inflammatory cascades. Following O₃ exposure, respiratory epithelial cells undergo direct oxidative damage, releasing ROS (e.g., superoxide anion O₂⁻, hydroxyl radical $\cdot$ OH, hydrogen peroxide H₂O₂), which then diffuse systemically through the circulatory system, triggering systemic oxidative stress [10]. Some studies suggest that high-concentration or long-term exposure to sulfur dioxide (SO₂) can induce oxidative stress in vascular endothelial cells through the production of metabolites such as sulfites and sulfates, activate the NF- κ B signaling pathway, and promote the release of inflammatory factors, leading to endothelial dysfunction [11]. Additionally, SO₂ exposure can reduce the activity of antioxidant enzymes such as glutathione peroxidase (GSH-Px) and superoxide dismutase (SOD), impairing the body's antioxidant capacity and contributing to vascular inflammation and heart disease [12]. Particulate matter (PM₁, PM_{2.5}, PM₁₀), as major components of ambient air pollution, exert multi-pathway and multi-target effects on the cardiovascular and cerebrovascular systems. Notably, smaller particle sizes correspond to stronger penetration and more significant harm. PM can be inhaled through the respiratory tract, cross the alveolar barrier into the circulatory system, and act directly on vascular endothelial cells or indirectly trigger systemic inflammatory

responses by inducing pulmonary inflammation [13]. Furthermore, PM can affect coagulation function by upregulating the expression of thromboxane A2 (TXA2) and von Willebrand factor (vWF), while inhibiting the production of prostacyclin (PGI2). This leads to enhanced platelet aggregation capacity, increased risk of thrombosis, and ultimately induces myocardial infarction and stroke [14].

At the same time, the application of machine learning technology in the study of the relationship between environmental pollutants and diseases has also been widely explored. Some studies have explored the predictive value of air pollutants and socio-economic indicators on the mortality of endocrine, nutritional and metabolic diseases in Italian provinces by using regression models such as gradient boosting and random forest combined with SHAP analysis. It is found that specific pollutants such as benzene are the key factors affecting mortality, which confirms the important influence of the interaction between environmental and socio-economic factors on health outcomes [15]. Another study constructed a machine learning model based on volatile organic compounds (VOCs) exposure data and demographic data to predict the risk of cardiovascular disease. The random forest model showed the best predictive performance. SHAP analysis further clarified that age and VOCs metabolite ATCA were the core predictors, and ATCA had a protective effect on cardiovascular disease, which provided a new idea for individualized risk prediction of cardiovascular disease combined with environmental exposure data. Although the existing studies have confirmed the application value of machine learning in this field, most of them have not carried out a comprehensive correlation analysis between various air pollutants and the three core cardiovascular and cerebrovascular adverse outcomes of hypertension, heart disease and stroke, and there are still research gaps [16].

This study integrates data from the China Health and Retirement Longitudinal Study (CHARLS) and the China High-resolution Air Pollutants (CHAP) dataset. It includes seven major air pollutants (PM1, PM2.5, PM10, NO2, SO₂, CO, O₃) and incorporates multi-dimensional covariates such as demographic characteristics, lifestyle habits, and underlying diseases. Through traditional statistical analyses and machine learning methods, this study systematically investigates the associations between long-term air pollution exposure and hypertension, heart disease, and stroke among middle-aged and elderly individuals, identifies key pollutants influencing adverse cardiovascular and cerebrovascular outcomes, and provides a scientific basis for formulating precise environmental health intervention strategies and CCVD prevention and control measures.

Materials and methods

Study population

Data for this study were obtained from the China Health and Retirement Longitudinal Study (CHARLS). The analysis was divided into two time periods based on research objectives: (1) AQI-related analysis: included follow-up survey data from 2015 to 2020; (2) Machine learning analysis: included follow-up survey data from 2013 to 2020.

Initiated in 2011, CHARLS is a large-scale interdisciplinary survey jointly implemented by Wuhan University and Peking University, funded by the National Natural Science Foundation of China. Its primary objective is to provide detailed data on the comprehensive living conditions, economic status, and health status of middle-aged and elderly individuals in China, to support public policy formulation and social science research development. CHARLS adopted a multi-stage stratified random sampling method, covering urban and rural areas across 31 provincial-level administrative units in China to construct a representative sample. The initial sampling included 105 communities and 150 townships, with approximately 150 residential units randomly selected within each community and township. Participants were aged 45 years and above. Data collection in CHARLS was conducted through face-to-face surveys, including interviews and physical measurements. The survey content covered multiple domains such as personal background information, household composition, economic status, household income and assets, labor force status, health-care utilization, retirement, and health conditions.

Inclusion criteria were: (1) aged ≥ 45 years; (2) complete data on demographic characteristics, lifestyle habits, health outcomes, and corresponding air pollutant exposure levels. Exclusion criteria were: (1) missing diagnostic information on adverse cardiovascular and cerebrovascular outcomes (hypertension, heart disease, stroke); (2) missing air quality-related data; (3) missing key covariates (e.g., age, gender, history of underlying diseases).

For the AQI-related analysis, a total of 20,990 participants were included: 20,990 for the hypertension analysis (7,609 hypertension patients, 13,381 non-patients), 20,990 for the heart disease analysis (3,940 heart disease patients, 17,050 non-patients), and 6,736 for the stroke analysis (475 stroke patients, 6,261 non-patients). For the machine learning analysis, 22,517 participants were included: 22,517 for the hypertension analysis, 22,517 for the heart disease analysis, and 7,312 for the stroke analysis (the actual sample size for analysis was reduced due to excessive missing data). Owing to excessive missing values in individual indicators (e.g., partial pollutant concentration monitoring data, incomplete covariate information), the complete case analysis was adopted for data

processing in this study, with samples containing missing data directly excluded.

Exposure data acquisition

Air Quality Index (AQI) data

The AQI primarily includes concentrations of 7 major air pollutants (PM₁, PM_{2.5}, PM₁₀, nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), ozone (O₃)). AQI data were obtained from the Science Data Bank database, which provides long-term, high-reliability environmental monitoring data in China, covering routine air quality monitoring indicators in major cities and regions nationwide. The data have undergone rigorous quality control and standardization, making them suitable for exposure quantification in epidemiological studies [17].

Annual average AQI data corresponding to the residential locations (at the city level) of each participant from 2015 to 2020 were extracted as indicators of individual long-term AQI exposure levels. For individual missing monitoring data, records were deleted to ensure the accuracy of exposure data.

Pollutant concentration data

Exposure factors in the machine learning analysis included concentrations of six major air pollutants (PM₁, PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃). Data were obtained from the China High Air Pollutants (CHAP) dataset (<https://weijing-rs.github.io/product.html>) [18, 18]. This dataset integrates multi-source satellite remote sensing data, ground-based monitoring station data, atmospheric reanalysis data, and model simulation results, optimized using an artificial intelligence algorithm based on the space–time extremely randomized trees (STET) model. The core principles and advantages of this algorithm are as follows: ① It leverages the integrated learning capability of random forests to mine complex non-linear correlations between pollutant concentrations and multi-dimensional influencing factors (meteorological conditions, topography, emission sources, etc.); ② It introduces spatiotemporal weight factors, optimizing the capture of spatial autocorrelation and temporal dynamic changes of pollutants through Haversine distance calculation and time radian difference adjustment; ③ It combines data correction and feature selection (Gini index) to reduce errors from satellite remote sensing data and interference from redundant variables. This algorithm enables spatially continuous estimation of pollutant concentrations with a 1 km high resolution, which has been verified in similar high-resolution atmospheric pollutant dataset construction with a coefficient of determination (R²) of over 0.89 and a root mean square error (RMSE) of less than 11 µg/m³. The CHAP dataset provides high-quality, high spatiotemporal resolution air pollutant exposure

concentration data in China, and has been widely used in research on air pollution and public health.

Annual average concentrations of PM₁, PM_{2.5}, O₃, NO₂, SO₂, and CO from 2013 to 2020 were calculated. The spatial distribution of these six air pollutants was matched with the residential addresses in the CHARLS database to estimate individual long-term exposure levels to the six air pollutants. Missing data were handled using the same method as for AQI data.

Calculation of city-level prevalence rates

Based on the residential city information of participants, the prevalence rates of adverse cardiovascular and cerebrovascular outcomes (hypertension, heart disease, stroke) were calculated by city. The formula was: Prevalence rate (%) of a specific outcome in a city = (Number of confirmed cases of the outcome in the city/Total number of participants in the city) × 100%. These rates were used to analyze the association between city-level prevalence rates and AQI.

Outcome indicator definitions

Adverse cardiovascular and cerebrovascular outcomes were the primary outcomes of the study, including hypertension, heart disease, and stroke. They were defined based on standardized questionnaires, physical examinations, and medical diagnostic records in the CHARLS database:

Hypertension: Confirmed if any of the following conditions were met: ① A history of hypertension with a previous medical diagnosis; ② Systolic blood pressure ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg in the current survey; ③ Current use of antihypertensive medications.

Heart disease: Comprehensively determined based on participants' self-report of "a previous medical diagnosis of heart diseases such as coronary heart disease, myocardial infarction, heart failure, or arrhythmia", combined with a history of hospitalization and relevant examination results (e.g., electrocardiogram, echocardiography).

Stroke: Confirmed based on participants' self-report of "a previous medical diagnosis of cerebrovascular diseases such as cerebral hemorrhage or cerebral infarction", combined with neurological examination records and hospitalization history.

All outcome indicators were defined as binary variables (1 = outcome occurred, 0 = outcome did not occur).

Covariate selection

Based on existing epidemiological evidence, potential confounding factors that may influence the association

between air pollution and adverse cardiovascular and cerebrovascular outcomes were selected as covariates, specifically including: age (continuous variable, years), gender (male/female), marital status (married/unmarried), residential type (urban/rural, using two classification standards: Residence Type 1 and Residence Type 2, based on geographic coding in the CHARLS database), household registration type (agricultural/non-agricultural), education level (primary school or below, junior high school, senior high school, college or above); alcohol consumption status (yes/no, defined as regular drinking in the past 12 months), smoking status (yes/no, defined as current smoking or having a long-term smoking history), physical activity (yes/no, defined as engaging in moderate-intensity or higher physical activity at least once a week); dyslipidemia (yes/no, based on total cholesterol ≥ 5.2 mmol/L, triglycerides ≥ 1.7 mmol/L, high-density lipoprotein cholesterol < 1.04 mmol/L, low-density lipoprotein cholesterol ≥ 3.4 mmol/L, or a previous diagnosis of dyslipidemia), history of hospitalization (yes/no, hospitalization due to illness in the past 12 months), history of other chronic diseases (yes/no, chronic diseases other than CCVDs, such as diabetes mellitus, chronic obstructive pulmonary disease).

Data analysis methods

All data analyses were performed using R software version 4.2.1 (R Foundation for Statistical Computing, Vienna, Austria), with a significance level of $\alpha = 0.05$ (two-tailed test).

Descriptive statistical analysis

Descriptive statistical methods were used to present the baseline characteristics of the study population. Participants were divided into case groups and control groups based on the occurrence of adverse cardiovascular and cerebrovascular outcomes (hypertension, heart disease, stroke). Continuous variables (e.g., age, AQI, pollutant concentrations, blood pressure) were expressed as mean $\pm$ standard deviation (Mean $\pm$ SD), and comparisons between groups were performed using independent-samples t-tests. Categorical variables (e.g., gender, marital status, alcohol consumption status) were expressed as count (percentage, n, %), and comparisons between groups were performed using Pearson's chi-square test or Fisher's exact test (when the expected

frequency < 5). Baseline characteristic tables for hypertension, heart disease, and stroke were constructed separately (Tables S1–S3, Tables S4–S6) to comprehensively display the basic characteristics of the study population and differences between groups [19].

Single factor linear regression analysis

Taking the city as the analysis unit, we first confirmed that the continuous variables included in this analysis conformed to the normal distribution, satisfying the basic application assumptions of the linear regression model. On this basis, this study adopted a single-factor linear regression model to investigate the potential linear correlation between the prevalence of adverse cardiovascular and cerebrovascular outcomes (including hypertension, various types of heart disease, and stroke) and the annual average Air Quality Index (AQI) of cities spanning the 2015–2020 period. In this regression model, the city-level prevalence of adverse cardiovascular and cerebrovascular outcomes was set as the continuous dependent variable, while the annual average AQI of each corresponding city was defined as the independent variable.

To systematically and comprehensively evaluate the linear association between the two indicators, relevant statistical parameters were calculated via linear regression analysis, including the regression coefficient (β), standard error (SE), 95% confidence interval (95% CI) for the regression coefficient, and the corresponding *P* value. These core statistical indicators were applied to test whether a statistically significant linear correlation existed between the city-level prevalence of cardiovascular and cerebrovascular diseases and urban annual average AQI. The detailed results of this single-factor linear regression analysis are systematically presented and summarized in Table 1 for intuitive reference [20].

Multivariate logistic regression analysis

Multivariate logistic regression models were used to analyze the association between AQI exposure from 2015 to 2020 and adverse cardiovascular and cerebrovascular outcomes (hypertension, heart disease, stroke) at the individual level, to evaluate the impact of AQI on the risk of individual CCVD outcomes. Three progressively adjusted regression models were constructed to control for potential confounding factors:

Model 1 (Non-adjusted): Only included AQI as the exposure factor, without adjusting for any covariates. Model 2 (Adjust I, partially adjusted): Adjusted for two core demographic covariates: age (continuous variable) and gender (categorical variable). Model 3 (Adjust II, fully adjusted): Based on Model 2, further adjusted for all potential covariates including marital status, residential type, household registration type, education level, alcohol consumption status, smoking status, physical activity

Table 1 The relationship between AQI exposure and the prevalence of cardiovascular and cerebrovascular adverse outcomes

Exposure	Hypertension	Heart disease	Stroke
AQI	$\beta = 0.0006, 95\%$ CI: (0.0001– 0.0010), $P = 0.0125$	$\beta = 0.0008,$ 95% CI: (0.0003, 0.0013), $P = 0.0147$	$\beta = 0.0005, 95\%$ CI: (0.0002– 0.0008), $P = 0.0248$

status, dyslipidemia, history of hospitalization, and history of other chronic diseases. All regression results were expressed as odds ratios (OR) and 95% confidence intervals (95% CI). A P -value < 0.05 was considered statistically significant (Table 2) [21].

Restricted Cubic Spline (RCS) analysis

To explore the potential non-linear dose–response relationship between AQI exposure and adverse cardiovascular and cerebrovascular outcomes, restricted cubic spline (RCS) analysis was used to extend the results of the multivariate logistic regression models. Separate RCS models were constructed for hypertension, heart disease, and stroke based on the three progressively adjusted models (non-adjusted, partially adjusted, fully adjusted). Three knots were set based on the 25th, 50th, and 75th percentiles of the AQI distribution, with the median AQI as the reference value. Dose–response curves were plotted (Fig. 1). Specifically: Fig. 1A–C: RCS curves for hypertension in the non-adjusted, partially adjusted, and fully adjusted models, respectively. Figure 1D–F: RCS curves for heart disease in the non-adjusted, partially adjusted, and fully adjusted models, respectively. Figure 1G–I: RCS curves for stroke in the non-adjusted, partially adjusted, and fully adjusted models, respectively.

The linear or non-linear characteristics of the associations were determined. A significant non-linear trend with a P -value < 0.05 indicated a non-linear association between AQI and the outcome.

Machine learning analysis

Based on data from 2013 to 2020, predictive models were constructed for hypertension, heart disease, and stroke, respectively, using six commonly used binary classification machine learning algorithms: XGBoost (Extreme Gradient Boosting), Random Forest, Decision Tree, Gradient Boosting Decision Tree (GBDT), Multi-Layer Perceptron (MLP), and K-Nearest Neighbors (KNN) [22, 23]. These algorithms were selected for their mature theoretical foundation and diverse technical characteristics, which can adapt to the high-dimensional, heterogeneous data features of this study (including multiple air pollutants and multi-dimensional covariates), fully capture the nonlinear correlations and interaction effects between variables, and avoid the limitations of traditional linear statistical methods. Meanwhile, the multi-algorithm comparison verification strategy can effectively avoid

the inherent bias of a single algorithm, and ensure the robustness and reliability of the model prediction performance and key pollutant identification results.

The core hyperparameters of each algorithm for the three disease outcomes are detailed in Supplementary Materials, with consistent settings for most parameters and disease-specific adjustments for key parameters (Supplementary Materials 1.1).

The training of machine learning models in this study is based on the annual average concentration data of air pollutants in the residence of the study subjects from 2013 to 2020, as well as the average data of covariates such as demography and lifestyle.

All analysis data were randomly divided into a training set (for model training and parameter optimization) and a test set (for model performance validation) at an 8:2 ratio. To mitigate the potential impact of category imbalance between the case and control groups on model performance, we strictly ensured that the sample ratio of the two groups in the training and test sets was consistent with that of the original dataset during the division process. Five-fold cross-validation was used for internal validation of the training set to avoid model overfitting and ensure the robustness of the results. In the modeling process of all six binary classification algorithms, we set class weights inversely proportional to the category distribution, assigning higher weights to the minority heart disease case group to enhance the model's learning of minority class features and reduce prediction bias caused by the dominant majority class. In addition, the model performance was not evaluated by a single accuracy index, but comprehensively assessed by combining AUC, sensitivity, specificity, calibration curves and other multi-dimensional indicators, with a focus on the model's recognition ability for the minority class to ensure the reliability of the prediction results. The input variables of the models included concentrations of pollutants (PM₁, PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃) and all covariates. The output variables were adverse cardiovascular and cerebrovascular outcomes (hypertension/heart disease/stroke: 1 = occurred, 0 = did not occur).

The following indicators were used to comprehensively evaluate the predictive performance of each machine learning model. Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC), the AUC ranges from 0 to 1, with a value closer to 1 indicating better predictive performance. An AUC ≥ 0.7 suggests

Table 2 Multivariate logistic regression analysis results of AQI exposure and adverse cardiovascular and cerebrovascular outcomes

Exposure	AQI	AQI	AQI
Outcome	Hypertension	Heart Disease	Stroke
Non-adjusted	OR = 1.003, 95% CI: (1.002, 1.004), $P < 0.001$	OR = 1.006, 95% CI: (1.005, 1.007), $P < 0.001$	OR = 1.002, 95% CI: (0.999, 1.005), $P = 0.2746$
Adjust I	OR = 1.004, 95% CI: (1.003, 1.005), $P < 0.001$	OR = 1.007, 95% CI: (1.006, 1.008), $P < 0.001$	OR = 1.003, 95% CI: (1.000, 1.006), $P = 0.0660$
Adjust II	OR = 1.003, 95% CI: (1.002, 1.004), $P < 0.001$	OR = 1.006, 95% CI: (1.005, 1.007), $P < 0.001$	OR = 1.004, 95% CI: (1.001, 1.007), $P = 0.0433$

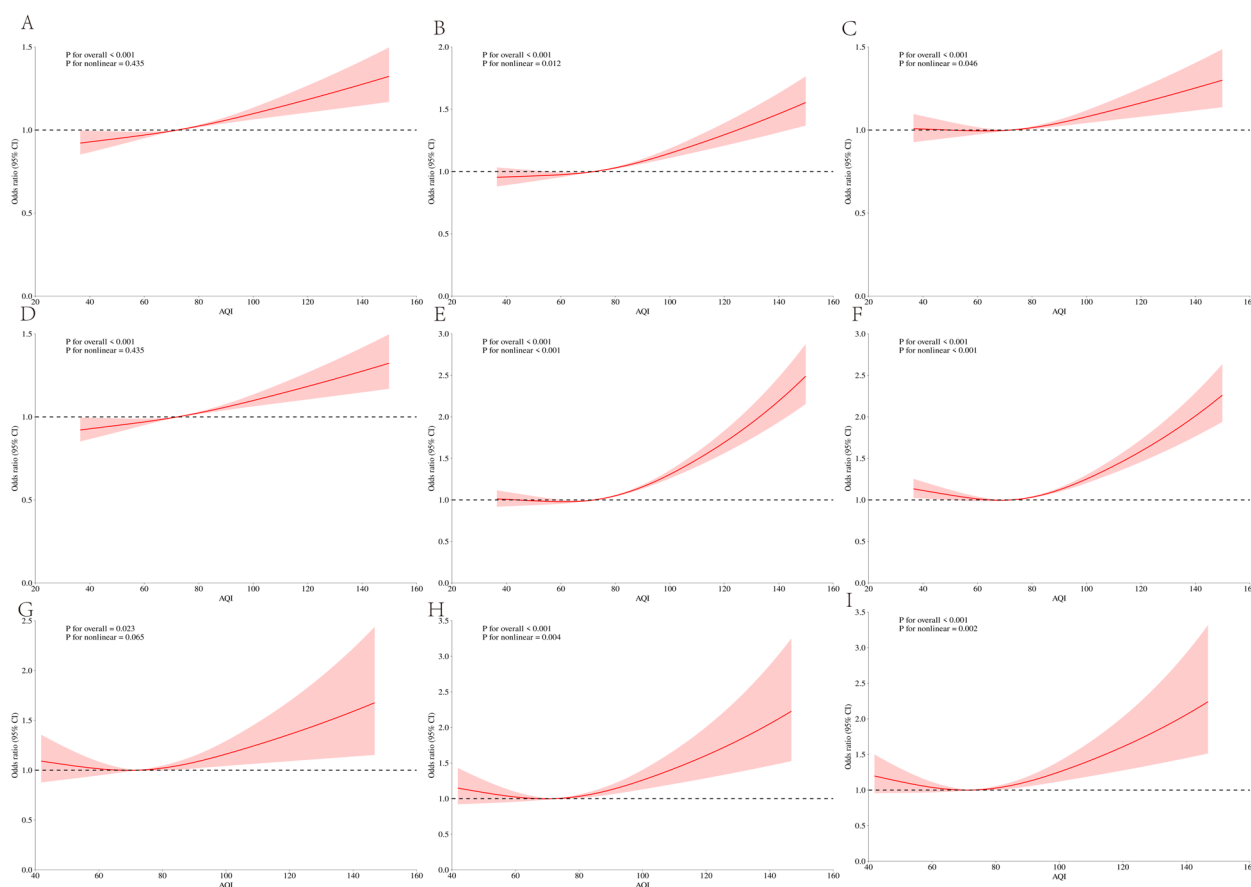


Fig. 1 Restricted cubic spline (RCS) analysis curve of air pollution AQI exposure and the risk of hypertension, heart disease and stroke in middle-aged and elderly people. In the figure, the horizontal axis represents the annual average air quality index (AQI) of the participants' residence from 2015 to 2020, and the vertical axis represents the risk ratio (OR) of cardiovascular and cerebrovascular adverse outcomes. The median of AQI is used as the reference value (OR=1.0), and the gray area represents the 95% confidence interval (95% CI). **A–C** were the unadjusted model, partially adjusted model (adjusted age, gender) and fully adjusted model (adjusted age, gender, marital status, type of residence and all other covariates) of hypertension. **D–F** were unadjusted model, partially adjusted model and fully adjusted model of heart disease, respectively. **G–I** were unadjusted model, partially adjusted model and fully adjusted model of stroke, respectively. In the figure, 'P for overall' represents the overall statistical significance of the association between AQI and outcome, and 'P for nonlinear' represents the nonlinear test results of the association between the two; if 95% CI did not include a reference line (OR=1.0), it suggested that there was a statistical difference between the risk of outcome and the reference value at this AQI level

good discriminative ability of the model. By comparing the model-predicted probability of outcome occurrence with the actual observed frequency of outcome occurrence, the calibration degree of the model was evaluated. A curve closer to the diagonal (Perfectly Calibrated) indicates better calibration performance. Used to display the AUC values and standard deviations (SD) of each model in the training set and test set, to visually compare the performance differences between different algorithms.

Finally, Shapley Additive Explanations (SHAP) values were used to analyze the contribution of each input variable (especially pollutants) to the model output, to identify the key pollutants associated with adverse cardiovascular and cerebrovascular outcomes. SHAP values quantify the importance of features by calculating the marginal contribution of each feature to the prediction result. A larger mean absolute SHAP value (mean

(|SHAP value|)) indicates a higher weight of the feature's impact on the outcome. SHAP summary plots and importance ranking plots were used to visually display the relative importance of NO₂ and other pollutants in influencing adverse cardiovascular and cerebrovascular outcomes [24, 25].

Results

Baseline characteristics of the study population

Hypertension-related baseline characteristics (Tables S1, S4)

Among the 20,990 participants in the 2015–2020 AQI-related analysis, 7,609 (36.2%) were hypertension patients and 13,381 (63.8%) were non-hypertension patients. The case group was significantly older than the control group (64.4 ± 10.2 years vs. 59.7 ± 9.9 years, $P < 0.001$), and the AQI level in the case group (80.7 ± 22.9) was significantly higher than that in the control group (79.0 ± 22.5 ,

$P < 0.001$). For categorical variables, the proportions of unmarried individuals (18.6% vs. 11.9%), dyslipidemia (36.2% vs. 12.9%), history of hospitalization (22.3% vs. 12.8%), and physical inactivity (13.3% vs. 9.5%) in the case group were significantly higher than those in the control group, while the proportions of alcohol consumption (31.4% vs. 36.4%) and smoking (23.6% vs. 28.6%) were significantly lower (all $P < 0.001$). There was no significant difference in gender distribution between the two groups ($P = 0.55$).

Among the 22,517 participants in the 2013–2020 machine learning analysis, the concentrations of PM1 ($23.26 \pm 9.15 \mu\text{g}/\text{m}^3$ vs. $22.71 \pm 8.98 \mu\text{g}/\text{m}^3$), PM2.5 ($41.16 \pm 16.40 \mu\text{g}/\text{m}^3$ vs. $40.23 \pm 16.11 \mu\text{g}/\text{m}^3$), PM10 ($93.61 \pm 12.47 \mu\text{g}/\text{m}^3$ vs. $93.82 \pm 8.71 \mu\text{g}/\text{m}^3$), CO ($0.92 \pm 0.27 \text{ mg}/\text{m}^3$ vs. $0.90 \pm 0.25 \text{ mg}/\text{m}^3$), and SO₂ ($17.28 \pm 10.90 \mu\text{g}/\text{m}^3$ vs. $16.68 \pm 10.59 \mu\text{g}/\text{m}^3$) in the hypertension case group were significantly higher than those in the control group (all $P < 0.001$). The difference in O₃ concentration was also statistically significant ($93.61 \pm 12.47 \mu\text{g}/\text{m}^3$ vs. $93.25 \pm 12.38 \mu\text{g}/\text{m}^3$, $P = 0.042$).

Heart disease-related baseline characteristics (Tables S2, S5)

Among the 20,990 participants in the 2015–2020 AQI-related analysis, 3,940 (18.8%) were heart disease patients and 17,050 (81.2%) were non-heart disease patients. The case group was significantly older than the control group (63.8 ± 10.2 years vs. 60.9 ± 10.1 years, $P < 0.001$), and the AQI level in the case group (80.5 ± 22.5) was significantly higher than that in the control group (79.4 ± 22.6 , $P = 0.002$). In terms of gender, the proportion of males in the case group (51.3%) was significantly higher than that in the control group (46.7%, $P = 0.018$). Regarding marital status, the proportion of married individuals in the case group (88.7%) was higher than that in the control group (85.0%, $P = 0.003$).

For residential type, the proportions of rural residents in the case group (Residence Type 1: 68.6% vs. 56.3%; Residence Type 2: 83.1% vs. 73.5%) were significantly higher than those in the control group (all $P < 0.001$). Among health-related indicators, the proportions of dyslipidemia (41.1% vs. 16.8%), history of hospitalization (33.7% vs. 12.2%), and physical inactivity (12.2% vs. 10.6%) in the case group were significantly higher than those in the control group, while the proportions of alcohol consumption (38.0% vs. 33.8%) and smoking (36.3% vs. 24.6%) were significantly higher (all $P < 0.001$).

Among the 22,517 participants in the 2013–2020 machine learning analysis, the concentrations of PM1 ($23.30 \pm 9.18 \mu\text{g}/\text{m}^3$ vs. $22.84 \pm 9.03 \mu\text{g}/\text{m}^3$), PM2.5 ($41.47 \pm 16.38 \mu\text{g}/\text{m}^3$ vs. $40.43 \pm 16.19 \mu\text{g}/\text{m}^3$), PM10 ($93.65 \pm 12.45 \mu\text{g}/\text{m}^3$ vs. $93.34 \pm 12.41 \mu\text{g}/\text{m}^3$), NO₂ ($26.46 \pm 8.82 \mu\text{g}/\text{m}^3$ vs. $25.94 \pm 8.75 \mu\text{g}/\text{m}^3$), CO ($0.93 \pm 0.27 \text{ mg}/\text{m}^3$ vs. $0.90 \pm 0.26 \text{ mg}/\text{m}^3$), and SO₂

($17.38 \pm 10.92 \mu\text{g}/\text{m}^3$ vs. $16.84 \pm 10.69 \mu\text{g}/\text{m}^3$) in the heart disease case group were significantly higher than those in the control group (all $P < 0.001$). The difference in O₃ concentration was statistically significant ($93.65 \pm 12.45 \mu\text{g}/\text{m}^3$ vs. $93.34 \pm 12.41 \mu\text{g}/\text{m}^3$, $P = 0.038$).

Stroke-related baseline characteristics (Tables S3, S6)

Among the 6,736 participants in the 2015–2020 AQI-related analysis, 475 (7.05%) were stroke patients and 6,261 (92.95%) were non-stroke patients. The case group was significantly older than the control group (64.8 ± 9.8 years vs. 61.9 ± 10.4 years, $P < 0.001$). In terms of gender distribution, the proportion of males in the case group (55.2%) was significantly higher than that in the control group (47.0%, $P = 0.002$). Regarding marital status, the proportion of married individuals in the case group (87.8%) was higher than that in the control group (85.0%, $P = 0.015$).

For residential type, the proportion of rural residents in the case group (68.7% vs. 53.7%) was significantly higher than that in the control group ($P < 0.001$). Among health-related indicators, the proportions of alcohol consumption (57.5% vs. 38.0%), smoking (47.6% vs. 27.1%), and history of hospitalization (52.8% vs. 25.5%) in the case group were significantly higher than those in the control group (all $P < 0.001$). There was no significant difference in physical activity proportion between the two groups ($P = 0.189$). There was no significant difference in AQI level between the two groups (76.9 ± 21.2 vs. 75.8 ± 21.3 , $P = 0.368$).

Among the 7,312 participants in the 2013–2020 machine learning analysis, the concentrations of PM1 ($23.41 \pm 9.12 \mu\text{g}/\text{m}^3$ vs. $22.89 \pm 9.05 \mu\text{g}/\text{m}^3$), PM2.5 ($41.58 \pm 16.31 \mu\text{g}/\text{m}^3$ vs. $40.53 \pm 16.23 \mu\text{g}/\text{m}^3$), NO₂ ($26.59 \pm 8.83 \mu\text{g}/\text{m}^3$ vs. $25.99 \pm 8.76 \mu\text{g}/\text{m}^3$), CO ($0.94 \pm 0.28 \text{ mg}/\text{m}^3$ vs. $0.91 \pm 0.26 \text{ mg}/\text{m}^3$), and SO₂ ($17.48 \pm 10.95 \mu\text{g}/\text{m}^3$ vs. $16.89 \pm 10.71 \mu\text{g}/\text{m}^3$) in the stroke case group were significantly higher than those in the control group (all $P < 0.001$). There were no significant differences in PM10 and O₃ concentrations between the two groups (all $P > 0.05$).

Single factor linear regression analysis of city-level prevalence rates of adverse CCVD outcomes and AQI (Table 1)

Results of the single factor linear regression analysis showed that there were significant positive linear correlations between the city-level prevalence rates of adverse cardiovascular and cerebrovascular outcomes and the annual average Air Quality Index (AQI) across the study period from 2015 to 2020 (all $P < 0.05$). Prior to regression modeling, we confirmed that the relevant continuous variables conformed to the normal distribution,

meeting the basic application requirements of the linear regression model.

Specifically, for hypertension, each 1-unit increment in annual average AQI was linked to a statistically significant rise in the city-level hypertension prevalence rate, with a regression coefficient $\beta = 0.0006$ (95% confidence interval [95% CI]: 0.0001–0.0010, $P = 0.0125$). For various types of heart disease, each 1-unit increase in AQI corresponded to a significant elevation in the city-level heart disease prevalence rate, and the regression coefficient was calculated as $\beta = 0.0008$ (95% CI: 0.0003–0.0013, $P = 0.0147$). For stroke, each 1-unit rise in AQI was associated with a notable increase in the city-level stroke prevalence rate, with a regression coefficient $\beta = 0.0005$ (95% CI: 0.0002–0.0008, $P = 0.0248$).

Collectively, these findings demonstrate that higher city-level annual average AQI levels are correlated with increased population-level risks of adverse cardiovascular and cerebrovascular outcomes, and all the observed linear associations are statistically significant.

Multivariate logistic regression and RCS analyses of individual adverse CCVD outcomes and AQI

Multivariate logistic regression analysis (Table 2)

Results of the multivariate logistic regression analysis showed that after progressive adjustment for potential confounding factors, there were significant positive correlations between AQI exposure from 2015 to 2020 and the individual risks of adverse cardiovascular and cerebrovascular outcomes (hypertension, heart disease, stroke) (all $P < 0.05$), and the strength of the associations remained stable after adjustment.

For hypertension, in the non-adjusted model, each 1-unit increase in AQI was associated with a significant increase in the risk of hypertension (OR = 1.003, 95% CI: 1.002–1.004, $P < 0.001$). In the model adjusted for age and gender (Adjust I), the association remained significant with a slightly increased risk (OR = 1.004, 95% CI: 1.003–1.005, $P < 0.001$). After full adjustment for all covariates (Adjust II), the association remained significant: each 1-unit increase in AQI was associated with a 3% increase in the risk of hypertension (OR = 1.003, 95% CI: 1.002–1.004, $P < 0.001$).

For heart disease, in the non-adjusted model, each 1-unit increase in AQI was associated with a significant increase in the risk of heart disease (OR = 1.006, 95% CI: 1.005–1.007, $P < 0.001$). In the model adjusted for age and gender (Adjust I), the association was significantly strengthened (OR = 1.007, 95% CI: 1.006–1.008, $P < 0.001$). After full adjustment for all covariates (Adjust II), the association remained stable: each 1-unit increase in AQI was associated with a 6% increase in the risk of heart disease (OR = 1.006, 95% CI: 1.005–1.007, $P < 0.001$).

For stroke, in the non-adjusted model, the association between AQI and the risk of stroke was not statistically significant (OR = 1.002, 95% CI: 0.999–1.005, $P = 0.2746$). In the model adjusted for age and gender (Adjust I), the association was marginally significant (OR = 1.003, 95% CI: 1.000–1.006, $P = 0.0660$). After full adjustment for all covariates (Adjust II), the association reached statistical significance: each 1-unit increase in AQI was associated with a 4% increase in the risk of stroke (OR = 1.004, 95% CI: 1.001–1.007, $P = 0.0433$).

These results indicate that after controlling for individual-level confounding factors, AQI exposure from 2015 to 2020 remains an independent risk factor for the occurrence of adverse cardiovascular and cerebrovascular outcomes, with clear statistically significant associations.

Restricted Cubic Spline (RCS) analysis (Fig. 1)

Results of the RCS analysis further revealed the dose–response relationship between AQI exposure and adverse cardiovascular and cerebrovascular outcomes.

For hypertension, Fig. 1A (non-adjusted model) showed a significant association between AQI and the risk of hypertension (P for overall < 0.001) with a non-linear trend (P for nonlinear = 0.435). The risk increased progressively faster as AQI increased. Figure 1B (partially adjusted model) showed a persistent significant association (P for overall < 0.001) with a significant non-linear characteristic (P for nonlinear = 0.012). There was no significant shift in the curve trend after adjusting for age and gender. Figure 1C (fully adjusted model) showed a sustained significant association (P for overall < 0.001), but the non-linear characteristic disappeared (P for nonlinear = 0.046). The curve showed a stable linear increase, suggesting that after full control of confounding factors, the association between AQI and hypertension is primarily linear.

For heart disease, Fig. 1D (non-adjusted model) showed a significant association between AQI and the risk of heart disease (P for overall < 0.001), with no statistical significance in the non-linear test (P for nonlinear = 0.435). The curve showed an approximately linear increase. Figure 1E (partially adjusted model) showed a significant association (P for overall < 0.001) with a significant non-linear characteristic (P for nonlinear < 0.001). The risk increase rate accelerated significantly when AQI was at moderate to high levels (≥ 80). Figure 1F (fully adjusted model) showed persistent significant association and non-linear characteristic (both P for overall < 0.001 , P for nonlinear < 0.001). The curve showed a steep upward trend after AQI ≥ 80 , suggesting that moderate to high AQI levels have a more significant impact on heart disease risk.

For stroke, Fig. 1G (non-adjusted model) showed a significant association between AQI and the risk of stroke

(P for overall = 0.023), with a marginally significant non-linear test (P for nonlinear = 0.065). The risk increased slightly when AQI was at moderate levels (40–80). Figure 1H (partially adjusted model) showed a significantly strengthened association (P for overall < 0.001) with a statistically significant non-linear characteristic (P for nonlinear = 0.004). The risk showed a rapid upward trend after $\text{AQI} \geq 100$. Figure 1I (fully adjusted model) showed stable significant association and non-linear characteristic (P for overall < 0.001, P for nonlinear = 0.002). The curve deviated from the reference line ($\text{OR} = 1.0$) after $\text{AQI} \geq 80$, and the 95% CI did not include the reference line. This suggests that high-level AQI exposure significantly increases the risk of stroke, with a dose-dependent non-linear association.

The gray areas in all RCS curves represent the 95% confidence intervals. When the gray area does not include the reference line ($\text{OR} = 1.0$), it indicates that the risk of adverse cardiovascular and cerebrovascular outcomes at the corresponding AQI level is statistically different from the reference value (median AQI).

Machine learning model evaluation and key pollutant identification (based on 2013–2020 data)

Hypertension prediction models constructed using six machine learning algorithms all exhibited good predictive performance (Fig. 2A–C). The AUC range in the training set was 0.754–0.771, with the KNN algorithm showing the highest AUC (0.771), followed by the Random Forest algorithm (0.767). The AUC range in the test set was 0.748–0.764, with the KNN algorithm still performing the best (0.764), and the XGBoost and GBDT algorithms showing consistent performance ($\text{AUC} = 0.751$). The result of validation set is the core of model evaluation. Calibration curves (Fig. 2D) showed good consistency between the predicted probabilities and actual observed frequencies of all models, with curves close to the perfectly calibrated line, indicating reliable predictive accuracy of the models. Results of the SHAP value analysis (Fig. 2E–F) showed that among all input variables, NO_2 had the largest mean absolute SHAP value (mean ($|\text{SHAP value}|$) = 0.08), significantly higher than other pollutants. The SHAP summary plot intuitively showed that NO_2 had the highest contribution to hypertension prediction, with a wider distribution range of SHAP values and a significantly higher impact weight on model output than

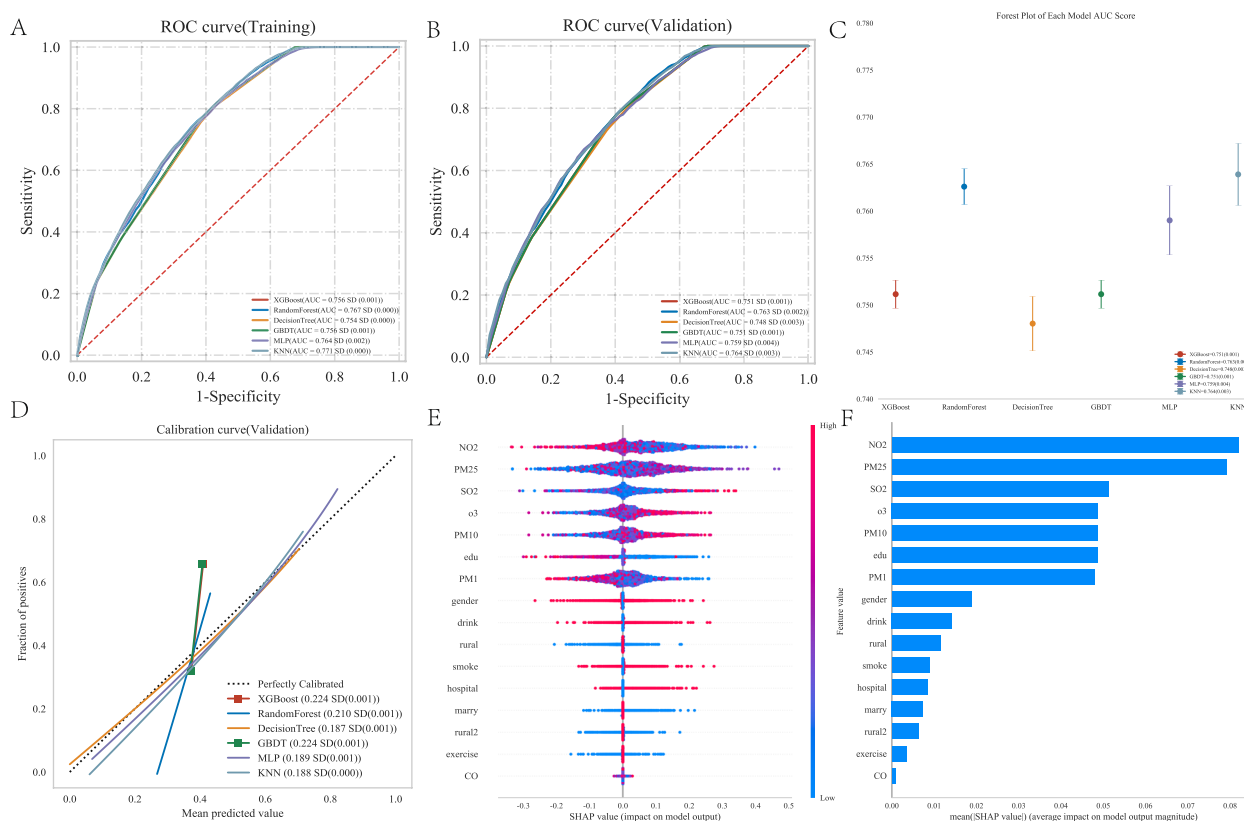


Fig. 2 Machine learning model evaluation of hypertension models incorporating all covariates. **A** The AUC results of the training set. **B** The AUC results of the test set. **C** The forest map of each machine learning algorithm. **D** Calibration curve of each machine learning algorithm. **E** The SHAP model summary results of the KNN algorithm. **F** The SHAP model importance ranking result of KNN algorithm

other pollutants, indicating that NO₂ is the most relevant pollutant to the incidence of hypertension.

Performance evaluation results of the heart disease prediction models showed (Fig. 3A–C) that the AUC range in the training set was 0.723–0.766, with the KNN algorithm performing the best (0.766), followed by the MLP algorithm (0.758). The AUC range in the test set was 0.720–0.760, with the KNN algorithm still being the best (0.760), and the MLP algorithm being the second best (0.755). The result of validation set is the core of model evaluation. Calibration curves (Fig. 3D) showed good calibration of all models, with curves close to the perfectly calibrated line, indicating reliable predictive accuracy of the models. Results of the SHAP value analysis (Fig. 3E–F) showed that NO₂ had a mean absolute SHAP value of 0.08, the highest among all pollutants. The SHAP importance ranking plot clearly showed that NO₂ ranked first in contribution to the heart disease prediction model, with a significantly higher marginal impact on model output than other pollutants, confirming that NO₂ is the most relevant pollutant to the incidence of heart disease.

For the stroke prediction models, the AUC range in the training set was 0.750–0.771, with the KNN algorithm

performing the best (0.771), followed by the Decision Tree algorithm (0.768). The AUC range in the test set was 0.735–0.757, with the KNN algorithm still being the best (0.757), and the Random Forest algorithm being the second best (0.755) (Fig. 4A–C). The result of validation set is the core of model evaluation. Calibration curves (Fig. 4D) showed good calibration of all models, with a high degree of overlap between the calibration curves and the perfectly calibrated line, indicating reliable predictive accuracy of the models for stroke. Results of the SHAP value analysis (Fig. 4E–F) showed that NO₂ had a mean absolute SHAP value of 0.035, significantly higher than other pollutants. The SHAP importance ranking plot clearly demonstrated the core role of NO₂ in stroke prediction, with a significantly higher impact weight than other pollutants, indicating that NO₂ is the most relevant pollutant to the incidence of stroke.

Detailed performance indicators (including accuracy, sensitivity, specificity, and other relevant metrics) of each model in the test set for hypertension, heart disease, and stroke prediction are provided in the Supplementary Materials for further reference (Supplementary Materials 1.2).

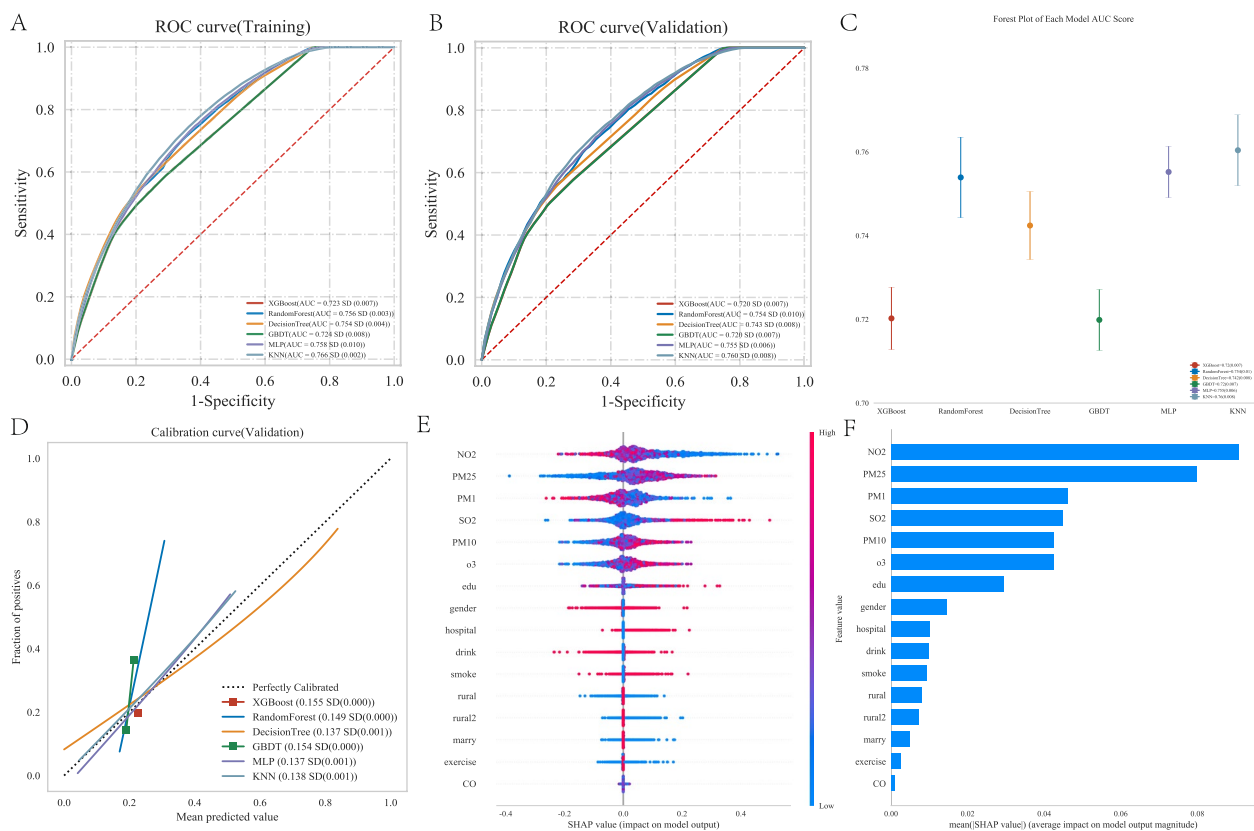


Fig. 3 Machine learning model evaluation of heart disease models incorporating all covariates. **A** The AUC results of the training set. **B** The AUC results of the test set. **C** The forest map of each machine learning algorithm. **D** Calibration curve of each machine learning algorithm. **E** Summary results of SHAP model of KNN algorithm. **F** The SHAP model importance ranking result of KNN algorithm

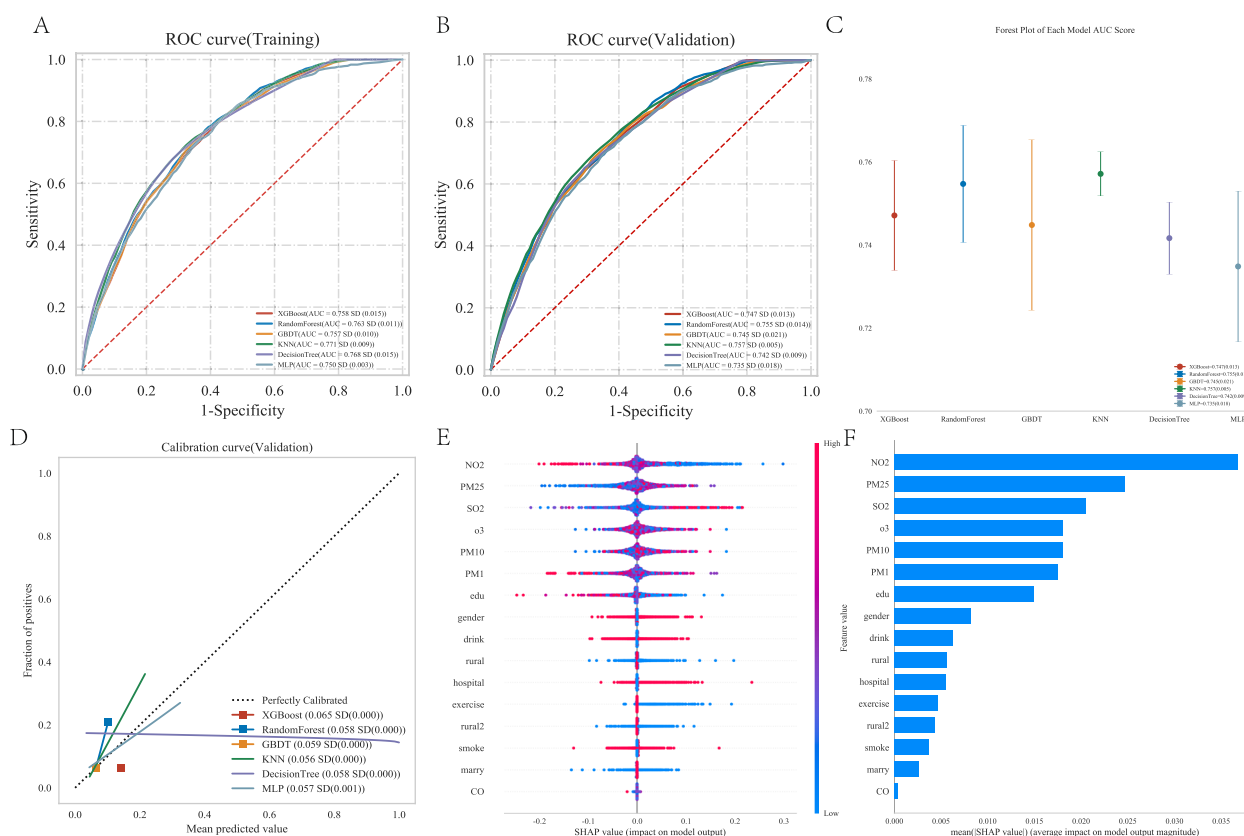


Fig. 4 Machine learning model evaluation of stroke models incorporating all covariates. **A** The AUC results of the training set. **B** The AUC results of the test set. **C** The forest map of each machine learning algorithm. **D** Calibration curve of each machine learning algorithm. **E** Summary results of SHAP model of KNN algorithm. **F** The SHAP model importance ranking result of KNN algorithm

Discussion

Adopting a progressive research logic that moves from population-level associations to individual-level effects, and finally to the identification of key pollutants, this study systematically revealed the mechanisms linking air pollution to adverse cardiovascular and cerebrovascular outcomes among middle-aged and elderly individuals. By integrating large-sample data from the CHARLS, Science Data Bank, and CHAP datasets, we provide multi-dimensional scientific evidence to support precise prevention and control strategies in environmental health.

Starting from a city-level perspective, our univariate linear regression analysis confirmed significant positive correlations between the annual average city-level AQI (2015–2020) and the prevalence rates of hypertension, heart disease, and stroke. Building on this macro-level association, individual-level multivariate logistic regression models demonstrated that AQI exposure remains an independent risk factor even after adjusting for comprehensive covariates (age, gender, lifestyle, underlying diseases). Specifically, each 1-unit increase in AQI was associated with a 3% increase in hypertension risk, 6% in heart disease risk, and 4% in stroke risk. Crucially, by

employing six machine learning algorithms combined with SHAP value analysis, we identified NO₂ as the most critical pollutant driving these three outcomes, exhibiting a mean absolute SHAP value significantly higher than other major pollutants such as PM_{2.5}, PM₁₀, and SO₂. This finding suggests that in the context of China's multi-pollutant environment, traffic-related emissions marked by NO₂ may pose a more immediate threat to cardiovascular health than previously emphasized particulate matter, warranting a shift in policy focus.

The inducing effect of NO₂ on hypertension is primarily mediated through vascular endothelial dysfunction and the disruption of vasodilation-constriction balance. NO₂-induced reactive oxygen species (ROS) can directly impair endothelial nitric oxide synthase (eNOS) activity, reducing the synthesis and release of nitric oxide (NO) [26, 27]. Since NO is essential for activating the sGC-cGMP pathway in vascular smooth muscle cells to promote relaxation, its reduction leads to sustained vasoconstriction and increased vascular resistance [28]. Furthermore, NO₂ activates the NF-κB signaling pathway in endothelial cells, upregulating vasoconstrictive substances such as endothelin-1 (ET-1) and angiotensin II (Ang II), which further exacerbate vascular contraction

and sodium retention via the PI3K/AKT and aldosterone pathways [27, 29]. These core mechanisms collectively drive the elevation of blood pressure.

Regarding heart disease, NO₂ exposure promotes cardiomyocyte injury and myocardial fibrosis. ROS generated by NO₂ inhalation can penetrate cardiomyocytes, causing mitochondrial damage, ATP depletion, and the release of pro-apoptotic factors (e.g., cytochrome C, AIF), thereby triggering caspase-mediated apoptosis and cell necrosis [30–32]. Concurrently, NO₂-induced inflammatory cytokines (e.g., TGF- β 1, IL-6) activate cardiac fibroblasts, leading to excessive collagen deposition and interstitial fibrosis via MAPK/JNK pathways [33, 34]. This structural remodeling disrupts electrical conduction and impairs both systolic and diastolic functions, progressing towards heart failure and arrhythmia.

For stroke, the pathogenesis involves blood–brain barrier (BBB) disruption and cerebrovascular fragility. NO₂-induced ROS oxidize tight junction proteins (occludin, ZO-1, claudin-5), increasing BBB permeability and allowing inflammatory cell infiltration, which triggers cerebral edema and ischemia [35]. Chronic exposure also promotes vascular wall inflammation and matrix metalloproteinase (MMP)-mediated degradation of the basement membrane, weakening vascular mechanical strength [36, 37]. Coupled with NO₂-induced platelet activation and microthrombi formation, these changes significantly elevate the risk of both hemorrhagic and ischemic stroke [38, 39]. While previous studies have detailed complex intracellular cascades, our results highlight that the convergence of oxidative stress, systemic inflammation, and structural vascular damage constitutes the primary pathway through which NO₂ exerts its clinical toxicity.

A comparative analysis with similar studies further underscores the novelty and significance of our findings. Most existing epidemiological research has focused on PM_{2.5} as the primary air pollutant linked to CCVDs: a large U.S. cohort study reported that each 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} was associated with a 16% higher risk of ischemic heart disease mortality and 14% higher stroke mortality [40], while a systematic review of PM_{2.5}-induced ischemic stroke mechanisms emphasized its roles in systemic inflammation, oxidative stress, and platelet activation [41]. A child cohort study also identified PM_{2.5}-related lipid metabolism disorders as mediators of hypertension risk [42]. However, these studies either overlooked NO₂ or treated it as a secondary pollutant. In contrast, our multi-pollutant geospatial analysis combined with machine learning and SHAP value quantification reveals that NO₂ exerts a more prominent impact on hypertension, heart disease, and stroke than PM_{2.5} and other pollutants. This aligns with a subset of studies that have highlighted NO₂'s unique

vascular toxicity—for example, NO₂ has been shown to directly disrupt vascular endothelial integrity and induce vasoconstriction independent of PM_{2.5}, but our study extends this evidence by confirming its primacy across three core CCVD outcomes in a large nationwide middle-aged and elderly population.

This finding carries profound implications for public health intervention. Traditionally, air pollution control strategies have prioritized PM_{2.5} reduction given its well-documented cardiovascular harms, but our results underscore that targeted NO₂ mitigation deserves integration into precision public health policies. Unlike PM_{2.5}—whose complex formation pathways include secondary aerosolization and broader environmental diffusion—NO₂ primarily stems from anthropogenic sources such as vehicle exhaust and industrial emissions, making it more localized and amenable to targeted control [43]. For public health practice, this translates to actionable priorities: in high-risk regions identified by our study (areas with elevated NO₂ and AQI), implementing stricter emission standards for diesel vehicles and industrial boilers could deliver more immediate reductions in CCVD risk than broad-spectrum PM_{2.5} control alone; integrating NO₂ monitoring data into CCVD risk stratification for middle-aged and elderly populations, such as recommending enhanced blood pressure screening and vascular function assessments for individuals residing in NO₂ hotspots; and complementing existing PM_{2.5}-focused early warning systems with NO₂-specific alerts, as the non-linear dose–response relationships we identified indicate that moderate-to-high NO₂ levels—even in the absence of extreme PM_{2.5} concentrations—may trigger accelerated CCVD risk. Furthermore, since NO₂ and PM_{2.5} often coexist in urban environments, synergistic control measures (e.g., promoting electric vehicles, optimizing industrial energy structures) can address both pollutants simultaneously, maximizing population-level cardiovascular protection [44].

From a policy perspective, prioritizing NO₂ control not only aligns with the precision of environmental health interventions but also promises substantial expected health benefits. Given the close link between NO₂ and the three core CCVD outcomes confirmed in this study, targeted emission reduction measures could directly translate to reduced population-level disease burden: modeling based on our risk estimates suggests that even a modest 10% reduction in ambient NO₂ concentrations might correlate with a 2%–3% decline in hypertension and heart disease prevalence among middle-aged and elderly populations, particularly in high-pollution urban areas [45]. Such benefits are both clinically meaningful and cost-effective, as NO₂ control strategies (e.g., optimizing urban traffic management, promoting clean energy in industrial production, and retrofitting

emission-control equipment for heavy-duty vehicles) often target localized, identifiable sources—enabling policymakers to allocate resources efficiently and track intervention effectiveness through routine air quality monitoring. Additionally, integrating NO₂ reduction into national and regional air quality improvement plans can complement existing PM_{2.5} control efforts, creating a synergistic effect that mitigates multiple pollutant-related health risks simultaneously. For aging societies facing a growing CCVD burden, these policy adjustments not only reduce direct healthcare costs associated with hypertension, heart disease, and stroke but also enhance population health resilience, ultimately contributing to the sustainability of public health systems.

This study's findings hold critical practical implications for the prevention of cardiovascular and cerebrovascular diseases (CCVDs) in middle-aged and elderly populations, especially in the context of environmental health intervention. First, the identification of NO₂ as the core harmful pollutant provides a precise target for air pollution control and CCVD primary prevention, suggesting that strengthening NO₂ emission reduction (e.g., from vehicle exhaust and industrial emissions) is essential to lower population-level CCVD risk. Second, the non-linear dose–response relationship reveals accelerated CCVD risk at moderate-to-high AQI levels, guiding the formulation of graded early warning and intervention strategies—prioritizing high-risk areas with elevated NO₂ and AQI for targeted health monitoring of middle-aged and elderly residents. Additionally, these results provide scientific evidence for cross-sector collaboration between environmental and health departments, supporting the design of integrated policies that combine air quality improvement with CCVD screening and intervention, ultimately reducing the disease burden of air pollution-related CCVDs in aging populations [46].

This nationwide multi-pollutant geospatial analysis identifies NO₂ as the key air pollutant associated with hypertension, heart disease, and stroke in middle-aged and elderly Chinese individuals, yet several limitations should be noted when interpreting the findings. First, the cross-sectional design, despite integrating 2013–2020 long-term exposure data, only reveals correlative rather than causal associations between air pollution and cardiovascular and cerebrovascular diseases (CCVDs), with unmeasured confounding factors (regional socioeconomic disparities) not fully excluded. Second, the study only assessed ambient outdoor air pollution and did not account for indoor pollution (e.g., cooking fumes, biomass combustion), leading to incomplete total exposure evaluation and potential bias in effect quantification. Third, no analysis of interactive or joint effects between pollutants was conducted; ambient air pollution is a complex mixture, and synergistic/additive effects between

NO₂ and other pollutants (PM_{2.5}, O₃) remain unelucidated, limiting understanding of multi-pollutant mixture impacts. Finally, CCVD outcomes relied on self-reported data from the CHARLS cohort, which may be subject to information bias due to varying medical cognition, and only three core outcomes were included, omitting other important CCVD sub-types (atrial fibrillation). These limitations highlight the need for prospective cohort studies, integrated indoor-outdoor exposure assessment, multi-pollutant interaction analysis, and objective outcome measures in future research to confirm causal links and refine targeted CCVD prevention strategies [46].

Conclusion

This study confirms that air pollution is an independent risk factor for hypertension, heart disease, and stroke in middle-aged and elderly individuals. Crucially, machine learning analysis identified NO₂ as the dominant pollutant, surpassing particulate matter in impact. NO₂ drives these outcomes primarily by inducing oxidative stress, impairing vascular endothelial function, and triggering systemic inflammation.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-026-27746-x>.

Supplementary Material 1.

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Authors' contributions

R. Zou, G. Li, S. Chen, Y. Liu and AISHAN.A take responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretations; they also drafted the article. J. Zhang take responsibility for the statistical analyses and interpretation of the data. S. Qiu take responsibility for the full text evaluation and guidance and performed the final approval of the version to be submitted. All authors read and approved the final manuscript.

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Data availability

Data is provided within the manuscript.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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